Conformance Test Report

IEC 61851-1, DIN 70121 and ISO 15118 (SLAC & V2G message) test cases — SECC under test (VTB)

Report generated: 2026-11-12 08:35:03

Overall: PASS

1 operator-accepted deviation(s)

System Under Test

Manufacturer	-
Model / type	-
Software / firmware	EVerest testing/main @ 9d7e0a4d266649c91fc1a77b2f35e9ffd4b9b24e
Serial / identification	-
Notes	Nightly CI, all SECC stacks Evse15118D20. SUT config: config-test-conformance-ac-dc100.yaml (one of the nightly's rig cycles, each with its own report)
Transfer mode	AC

Black-box DUT: the internal software stack is not observable by the test system; the identification above was entered by the test operator.

Declared capability profile (PICS/PIXIT)

Capability	PICS/PIXIT	Declared
Energy transfer mode	`PICS_CMN_CMN_ChargingMode`	**ac**
Payment options offered	`PIXIT_SECC_CMN_PaymentOption` / `PICS_CMN_CMN_IdentificationMode`	**ExternalPayment**
Certificate services offered	`PIXIT_SECC_CMN_CertificateHandling`	**none**
Certificate validity checked against current time	`PICS_SECC_CertificateValidationCheck`	**yes**
Tester CSMS is the DUT's OCPP backend	`PIXIT_SECC_TesterCSMS`	**yes**
Transport security accepted	`PIXIT_SECC_CMN_TLS`	**tcp**
Value-added service list (VAS)	`PIXIT_SECC_CMN_VAS == serviceDetail` / `PICS_SECC_VAS`	**no**
Internet access service	`PIXIT_SECC_CMN_InternetAccess == internetAccess`	**no**

Capability	PICS/PIXIT	Declared
Metering receipts requested	`PIXIT_SECC_CMN_ReceiptRequired`	**yes**
MeterInfo in charge-loop responses	`PIXIT_SECC_CMN_MeterInfo == meterInfo`	**yes**
MeteringConfirmation requested (ISO 15118-20)	`PICS_CMN_MeteringConfirmationRequested`	**no**
SalesTariff in schedules	`PIXIT_SECC_CMN_SalesTariff == salesTariff`	**no**
Control modes offered (ISO 15118-20)	`PICS_CMN_ControlMode`	**Dynamic**
Schedule initially unavailable (ISO 15118-20)	`PIXIT_SECC_ScheduleRetrievalSuspended`	**no**
DepartureTime considered	`PICS_SECC_CMN_DepartureTime`	**yes**
MaxEntriesSAScheduleTuple considered	`PICS_SECC_CMN_MaxEntriesSAScheduleTuple`	**yes**
Renegotiation supported	`PICS_CMN_CMN_Renegotiation`	**no**
CP State D (ventilation) during charging	`PICS_SECC_CMN_EvStateInChargeProgress == D`	**no**
Pause / resume supported	`PICS_SECC_CMN_Pause`	**no**
Welding detection supported	`PICS_SECC_DC_WeldingDetection`	**n/a (DC-only)**
ISO 15118-2 supported	`PICS_SECC_CMN_ISO15118_2`	**yes**
ISO 15118-20 supported	`PICS_SECC_CMN_ISO15118_20`	**no**
DIN SPEC 70121 supported (DC)	`PICS_SECC_DC_DIN70121`	**n/a (DC-only)**
DC energy transfer mode(s) advertised	`PIXIT_SECC_DC_SupportedEnergyTransferMode`	**n/a (DC-only)**
CableCheck answers Ongoing	`PIXIT_SECC_DC_CableCheckOngoing == ongoing`	**n/a (DC-only)**
Authorization answers Ongoing	`PIXIT_SECC_CMN_AuthorizationOngoing == ongoing`	**no**
ChargeParameterDiscovery answers Ongoing	`PIXIT_SECC_CMN_ChargeParameterDiscoveryOngoing == ongoing`	**no**
CM_VALIDATE (BCB-toggle validation)	`PIXIT_SECC_CMN_CmValidate == cmValidate`	**yes**
Architecture makes validation unnecessary	`PIXIT_SECC_CMN_ArchitectureValidationNotRequired`	**no**
CM_AMP_MAP supported	`PIXIT_CMN_CMN_CmAmpMap`	**no**

Capability	PICS/PIXIT	Declared
SECC initiates CM_AMP_MAP	PICS_CMN_CMN_InitiateCmAmpMap	**no**
Self-initiated wake-up from pause	PIXIT_CMN_CMN_WakeUp < PICS_CMN_CMN_WakeUp	**no**
AC connection-loss handling	PIXIT_SECC_AC_ConnectionLossHandling	**optionA**
AC CP duty cycle after plug-in	PIXIT_SECC_AC_InitialDutyCyle	**dc100**
EIM completion handled by the SUT	PICS_SECC_CMN_EIMDone	**beforePlugin, afterPlugin, duringSlac, v2gA**
IEC 61851-1 basic charging (AC)	PIXIT_SECC_IEC61851_1_BasicCharging	**yes**
Identification modes supported	PICS_CMN_CMN_IdentificationMode	**ExternalPayment**
Certificate services used	PIXIT_EVCC_CMN_CertificateHandling	**none**
Renegotiation supported	PICS_CMN_CMN_Renegotiation	**no**
Pause / resume (sleep mode) supported	PIXIT_EVCC_CMN_Pause	**no**
Optimises the charging profile against sales tariffs	PIXIT_EVCC_CMN_ChargingProfileOptimization	**unknown**
States MaxNumberOfPhasesInUse (AC)	PIXIT_EVCC_AC_MaxNumberOfPhasesInUse	**no**
Queries value-added services (ServiceDetail)	PIXIT_EVCC_CMN_VAS	**no**
Transport security requested	PIXIT_EVCC_CMN_TLS	**none_**
Welding detection performed (DC)	PIXIT_EVCC_DC_WeldingDetection	**weldingDetection**
Reaction to a Pmax=0 W schedule	PICS_EVCC_CMN_PmaxSchedulewithZeroPow	**noSleep**
SLAC validation (CM_VALIDATE / BCB toggle)	PIXIT_EVCC_CMN_CmValidate	**unknown**
AC connection-loss handling	PIXIT_EVCC_AC_ConnectionLossHandling	**optionA**
CM_AMP_MAP supported	PIXIT_CMN_CMN_CmAmpMap	**no**
EV initiates CM_AMP_MAP	PICS_CMN_CMN_InitiateCmAmpMap	**no**

Declared by the test operator (profile v1); the test-case selection of this run was derived from this declaration, mirroring the ISO 15118-4 ATS control part ([V2G4-018]/[V2G4-019]). Test cases the declaration marks not applicable were deselected by default; any that were explicitly force-run are annotated “declared N/A (operator force-run)” at their result.

Test families & variants

ISO 15118-3 SLAC (-5)	DUT (dut)
ISO 15118-2 message layer (-4)	DUT (dut)

Test Environment

Test suite version	-
Test suite git SHA	-
Test suite branch	-
Mode	HW (CHARGEBRIDGE, BLACK-BOX DUT)
Host	CI runner
Kernel	6.8.0-86-generic
PLC taps (EVSE / EV)	- / cb_plc_ev
MQTT broker	/tmp/everest-conformance-mqtt.sock:1883
EVerest prefix	-
Pause between TCs	5 s
EV BSP module id	ev_bsp

Totals

ISO 15118-3 SLAC (-5)

DUT (dut)
2 pass
0 fail
1 accepted fail
0 not run
0 inconclusive

ISO 15118-2 message layer (-4)

DUT (dut)
2 pass
0 fail
0 not run
0 inconclusive

Verdict semantics: expected failures (author-flagged known deviations / non-deterministic cases) are reported as FAIL EXPECTED — genuine reds that are not hidden and must be explicitly accepted by the operator. A FAIL (or FAIL EXPECTED) accepted by the operator is reported as ACCEPTED FAIL with the mandatory justification and does not count

against the overall verdict. Overall verdict is PASS only if no test case FAILs (or is left FAIL EXPECTED without acceptance) on any variant.

DUT code coverage (whole run, all configurations combined): 37.3 % of 71964 lines — see DUT code coverage.

Operator-accepted deviations

Test case	Justification
TC_SECC_AC_VTB_CmSlacParm_003	Operator-accepted deviation from ISO 15118-3 [V2G3-M06-08]: after a positive EIM the dc100 SUT goes from X1 straight to nominal PWM without the E/F step for T_step_EF. Kept on purpose for compatibility with EVs without HLC that follow plain IEC 61851-1 charging and are confused by the extra E/F step.

Summary matrix — ISO 15118-3 SLAC (-5)

Test case	DUT (dut)	
	Verdict	Duration
CmSlacParm		
TC_SECC_AC_VTB_CmSlacParm_003	ACCEPTED FAIL operator-accepted deviation	19.14 s
PLCLinkStatus		
TC_SECC_AC_VTB_PLCLinkStatus_009	PASS	6.72 s
TC_SECC_AC_VTB_PLCLinkStatus_011	PASS	12.28 s

Summary matrix — ISO 15118-2 message layer (-4)

Test case	DUT (dut)	
	Verdict	Duration
PaymentServiceSelection		
TC_SECC_CMN_VTB_PaymentServiceSelection_009	PASS	8.03 s
SDP		
TC_SECC_CMN_VTB_SDP_007	PASS	7.60 s

Test case details — ISO 15118-3 SLAC (-5)

TC_SECC_AC_VTB_CmSlacParm_003

TC_SECC_AC_VTB_CmSlacParm_003.

Test objective: Test System executes GoodCase procedure, indicates the initial CP State B waits for 100 % duty cycle and executes successful EIM process. Test System then checks that a nominal duty cycle can be detected after SUT changes CP State to E or F for T_step_EF. Furthermore a successful SLAC matching process shall be performed afterwards.

DUT (dut) ACCEPTED FAIL operator-accepted deviation 19.14 s

Known deviation: Operator-accepted deviation from ISO 15118-3 [V2G3-M06-08]: after a positive EIM the dc100 SUT goes from X1 straight to nominal PWM without the E/F step for T_step_EF. Kept on purpose for compatibility with EVs without HLC that follow plain IEC 61851-1 charging and are confused by the extra E/F step.

Operator-accepted failure: Operator-accepted deviation from ISO 15118-3 [V2G3-M06-08]: after a positive EIM the dc100 SUT goes from X1 straight to nominal PWM without the E/F step for T_step_EF. Kept on purpose for compatibility with EVs without HLC that follow plain IEC 61851-1 charging and are confused by the extra E/F step.

FAILURE REASON

tests/test_secc_ac_cm_slac_parm.py:132

```
TimeoutError: CP E/F (duty 0) step not observed before the window closed (last duty=26.666667938232422)
```

Measurement	Measured	Target / range	Result
initial CP duty cycle (dc100)	100.0 %	100 ± 1.5 %	✓

Failure detail (excerpt):

```
E   TimeoutError: CP E/F (duty 0) step not observed before the window closed (last duty=26.666667938232422)
```

TC_SECC_AC_VTB_PLCLinkStatus_009

> TC_SECC_AC_VTB_PLCLinkStatus_009.
>
> Test objective: Test System executes GoodCase procedure, initiates an EIM authorization during the SLAC process, indicates the key at the local PLC node and then checks that the data link connection is established within 'TT_match_join'. Test System then checks if the SUT has changed from 100 % initial duty cycle to a nominal duty cycle during the matching process by detecting X1(100 %) to X2(nominal) transition after successful EIM process during the matching process.

DUT (dut) PASS 6.72 s

Conformance check	Observed	Result
nominal duty applied within par_SECC_change_to_Nominal (3.0s) of the EIM	nominal duty 0.339s after EIM	✓
X2(nominal) duty period observed		✓
X1(100%) period observed before the nominal duty		✓
no 5% HLC duty period precedes the nominal duty (dc100 X1→X2(nominal) transition)		✓
CP duty valid throughout the dc100 X1→X2(nominal) transition chain	all samples within range	✓

Measurement	Measured	Target / range	Result
initial CP duty cycle (dc100)	100.0 %	100 ± 1.5 %	✓
CM_SLAC_PARM.CNF response time	2.4 ms	≤ 200 ms (TT_match_response)	✓
CM_ATTEN_CHAR.IND after sounding	420.8 ms	≤ 1200 ms (TT_EV_atten_results)	✓
CM_SLAC_MATCH.CNF response time	1.8 ms	≤ 200 ms (TT_match_response)	✓
CM_SET_KEY.CNF response time	1.1 ms	-	info
nominal duty after EIM (par_SECC_change_to_Nominal)	0.339 s	≤ 3.0 s	✓
nominal CP duty cycle after EIM (dc100 duringSlac)	26.7 %	[10.0, 96.0] %	✓

TC_SECC_AC_VTB_PLCLinkStatus_011

TC_SECC_AC_VTB_PLCLinkStatus_011.

Test objective: Test System executes GoodCase procedure (100 % duty cycle), indicates the key at the local PLC node and then checks that the data link connection is established within 'TT_match_join'. Afterwards Test System initiates a connection loss by setting a new key at the local node. Test System then checks that the SUT performs a CP State X1 to E/F transition. Furthermore Test System checks that the SUT applies CP State E or F for T_step_EF and will switch to CP State X1 or X2 afterwards.

DUT (dut) PASS 12.28 s

Conformance check	Observed	Result
SUT terminates the PLC link within TP_match_leave (1.25s) on connection loss (AC dc100)	left after 0.002s	✓
X1 → E/F directly on connection loss (no intervening X2 duty step) (AC dc100)	direct X1→E/F	✓
SUT sends no unexpected SLAC frame for 3.99318s	silent for 3.99318s	✓
CP E/F step ends within T_step_EF max on connection loss (AC dc100)		✓
CP E/F step within T_step_EF [4.0, 8.0]s on connection loss (AC dc100)	E/F step 4.016s	✓
CP duty after E/F step is a valid AC duty (3..7% or 10..100%) (AC dc100)	post-E/F duty 100.0%	✓

Measurement	Measured	Target / range	Result
initial CP duty cycle (dc100)	100.0 %	100 ± 1.5 %	✓
CM_SLAC_PARM.CNF response time	2.3 ms	≤ 200 ms (TT_match_response)	✓
CM_ATTEN_CHAR.IND after sounding	417.4 ms	≤ 1200 ms (TT_EV_atten_results)	✓
CM_SLAC_MATCH.CNF response time	1.0 ms	≤ 200 ms (TT_match_response)	✓
CM_SET_KEY.CNF response time	0.4 ms	-	info
CP duty at connection loss (X1 premise)	100.0 %	100 ± 1.5 %	✓
CM_SET_KEY.CNF response time	0.7 ms	-	info
left logical network on connection loss (AC dc100)	0.002 s	≤ 1.25 s	✓
T_step_EF on connection loss (AC dc100)	4.016 s	[4.0, 8.0] s	✓
CP duty after E/F step (AC: 3..7% or 10..100%) (AC dc100)	100.0 %	-	info
CM_SET_KEY.CNF response time	0.6 ms	-	info

Test case details — ISO 15118-2 message layer (-4)

TC_SECC_CMN_VTB_PaymentServiceSelection_009

> TC_SECC_CMN_VTB_PaymentServiceSelection_009.
>
> Test objective: GoodCase; Test System sends PaymentServiceSelectionReq with SelectedPaymentOption Contract and a valid ServiceID while the SUT is configured for externalPayment/eIM, and checks the SUT replies with the current SessionID, FAILED_PaymentSelectionInvalid, then closes TCP within par_CMN_TCP_Connection_Termination_Timeout.

DUT (dut) PASS 8.03 s

Conformance check	Observed	Result
Response echoes the current SessionID		✓
Response echoes the current SessionID		✓
ResponseCode = FAILED_PaymentSelectionInvalid	got 'FAILED_PaymentSelectionInvalid'	✓
SUT closes the TCP connection after the error response		✓

Measurement	Measured	Target / range	Result
CM_SLAC_PARM.CNF response time	1.1 ms	≤ 200 ms (TT_match_response)	✓
CM_ATTEN_CHAR.IND after sounding	411.5 ms	≤ 1200 ms (TT_EV_atten_results)	✓
CM_SLAC_MATCH.CNF response time	0.7 ms	≤ 200 ms (TT_match_response)	✓
CM_SET_KEY.CNF response time	0.2 ms	-	info
supportedAppProtocolRes response time	100.6 ms	≤ 2000 ms (V2G_EVCC_Msg_Timeout)	✓
SessionSetupRes response time	101.6 ms	≤ 2000 ms (V2G_EVCC_Msg_Timeout)	✓
ServiceDiscoveryRes response time	101.6 ms	≤ 2000 ms (V2G_EVCC_Msg_Timeout)	✓
PaymentServiceSelectionRes response time	101.4 ms	≤ 2000 ms (V2G_EVCC_Msg_Timeout)	✓
TCP connection close after failure/ timeout	0.1 ms	≤ 5000 ms (CMN_TCP_Connection_Termination_Timeout)	✓

TC_SECC_CMN_VTB_SDP_007

TC_SECC_CMN_VTB_SDP_007.

Test objective: GoodCase (eIM / no-TLS); the Test System sends an SDP request with Security=0×00 (TLS requested) and TransportProtocol=0×00, and checks the SUT -- which offers only no-transport-security -- answers an SDP response announcing Security=0×10 (no TLS), TransportProtocol=0×00 and a valid port + IP. This is the transport-security-negotiation (mismatch) case that matters for the TCP-only SUT: a TLS-requesting EV must be told to fall back to plain TCP.

DUT (dut) PASS 7.60 s

Conformance check	Observed	Result
SDP response TransportProtocol = 0×00 (TCP)	got 0×00	✓
SUT does not echo the requested Security=0×00 (TLS) back to the EV		✓
SDP response Security = 0×10 (no-TLS fallback)	got 0×10	✓
SDP response carries a valid port + IP	port=50000 addr=fe80::7004:9cff:fe51:e402	✓

Measurement	Measured	Target / range	Result
CM_SLAC_PARM.CNF response time	0.8 ms	≤ 200 ms (TT_match_response)	✓
CM_ATTEN_CHAR.IND after sounding	409.8 ms	≤ 1200 ms (TT_EV_atten_results)	✓
CM_SLAC_MATCH.CNF response time	0.8 ms	≤ 200 ms (TT_match_response)	✓
CM_SET_KEY.CNF response time	0.2 ms	-	info
SDP advertised TCP port	50000	-	info

DUT code coverage

Line coverage of the EVerest charger code reached by the test cases of the **whole run, all SUT configurations combined** (config-test-conformance-dc-full-ocpp.yaml, config-test-conformance-ac-basic.yaml, config-test-conformance-ac-full-ocpp.yaml, config-test-conformance-ac-dc100.yaml): a line counts as covered if any test case of any configuration executed it, so the figures are the same in every configuration's report. Measured with gcov on an instrumented build of EVerest 9d7e0a4d2666; the test cases themselves do not observe it.

Component	Files	Lines	Covered	Line %	Function %
EvseManager	42	5757	3886	67.5	74.4
Evse15118D20 (module)	20	2331	1001	42.9	54.6
libiso15118 DIN 70121	48	1932	1367	70.8	75.1
libiso15118 ISO 15118-2	72	4019	2445	60.8	68.3
libiso15118 ISO 15118-20	107	10098	2998	29.7	34.7
libiso15118 common	49	3113	2061	66.2	73.1
EvseSlac + libslac	53	3865	1736	44.9	51.9
OCPPmulti + libocpp 2.x	282	35619	9272	26.0	28.2
EvseSecurity + evse_security	22	3010	963	32.0	40.6
Auth	12	1348	536	39.8	58.0
Energy management	19	872	606	69.5	86.1
Total (charger code under test)	**726**	**71964**	**26871**	**37.3**	**41.6**

Not counted in the total

Rig scaffolding, framework, EXI codecs and code the conformance test cases do not target (e.g. the OCPP 1.6 stack); listed for completeness.

Component	Files	Lines	Covered	Line %
libocpp 1.6	106	12673	0	0.0
EXI codecs (cbv2g)	37	112399	10039	8.9
Framework	64	8416	2379	28.3
Rig: simulators, API bridges, test modules	38	990	453	45.8
Other EVerest code	242	26602	3620	13.6

Largest gaps

Files of the charger code under test with the most unexecuted lines.

File	Component	Lines missed	Line %
`lib/everest/ocpp/lib/ocpp/v2/ocpp_types.cpp`	OCPPmulti + libocpp 2.x	3150	15.8

File	Component	Lines missed	Line %
lib/everest/ocpp/lib/ocpp/v2/ocpp_enums.cpp	OCPPmulti + libocpp 2.x	1688	18.8
lib/everest/ocpp/include/ocpp/common/call_types.hpp	OCPPmulti + libocpp 2.x	1156	16.2
lib/everest/ocpp_module_common/src/conversions.cpp	OCPPmulti + libocpp 2.x	972	22.0
lib/everest/evse_security/lib/evse_security/evse_security.cpp	EvseSecurity + evse_security	822	30.8
modules/EVSE/OCPPmulti/generic_ocpp.cpp	OCPPmulti + libocpp 2.x	785	42.7
lib/everest/ocpp/include/ocpp/common/message_queue.hpp	OCPPmulti + libocpp 2.x	756	24.2
lib/everest/ocpp/include/ocpp/common/types.hpp	OCPPmulti + libocpp 2.x	727	1.0
modules/EVSE/OCPPmulti/v16_chargepoint.cpp	OCPPmulti + libocpp 2.x	685	0.0
lib/everest/ocpp/lib/ocpp/v21/functional_blocks/der_control.cpp	OCPPmulti + libocpp 2.x	652	0.0
lib/everest/ocpp/lib/ocpp/common/types.cpp	OCPPmulti + libocpp 2.x	626	7.3
lib/everest/iso15118/src/iso15118/message/ac_der_sae_charge_loop.cpp	libiso15118 ISO 15118-20	594	0.0
lib/everest/ocpp/lib/ocpp/v2/init_device_model_db.cpp	OCPPmulti + libocpp 2.x	544	31.5
modules/EVSE/EvseManager/EvseManager.cpp	EvseManager	544	70.5
lib/everest/iso15118/src/iso15118/d2/crypto.cpp	libiso15118 ISO 15118-2	530	24.8

The line-by-line report (every directory and file, with the unexecuted lines marked) is published next to this report on the run page: https://pionix-conformance-bot.github.io/Himalayas/reports/20261112T083454Z/coverage/index.html. gcovr 8.6, generated 2026-11-12 08:34 UTC.

Appendix

Codec self-tests

No codec self-tests were included in this run.

Toolchain / report metadata

pytest version	unknown
Report created	2026-11-12 08:33:04
pytest exit code	0
Report generator	Pionix conformance tester report generator
Generation timestamp	2026-11-12 08:35:03